# NEXUS-AI: FEATURE AUDIT & GAP RECONCILIATION

Automated Campus Research Project Lifecycle Management & Collaborative IP Filing Dashboard

*Comprehensive Verification: What is Implemented, What is Missing, and What is Extra / Advanced*

**VERIFIED PROJECT AUTHORS / CANDIDATES:**

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## 1. EXECUTIVE CONFIRMATION: DID WE IMPLEMENT ALL CONTENTS?

YES, the core functional contents claimed in the project report are comprehensively implemented and running live. The portal successfully links Students, Faculty, Academic Office, and IP Cell across 11 routes, tracks weekly milestone progress, executes local AI novelty scoring via Ollama (Llama 3.2) with fallback heuristics, and synthesizes Indian Patent Office Form 2 patent specifications. However, there are specific implementation gaps (mostly around security hardening, candidate names, and email triggers) alongside substantial EXTRA features that elevate the platform beyond the report.

| 28 / 32 Built | 4 Key Gaps | 7 Advanced Built |
|---|---|---|
| Core Report Features | Missing / Incomplete Gaps | Extra / Bonus Features |

## 2. WHAT IS MISSING / INCOMPLETE (GAPS TO RECTIFY)

**[MISSING - HIGH] 1. Team Candidate Names in Seed Data & Demo Logins:** The database seeds mock students ('Aarav Sharma', 'Ananya Patel', 'Rohan Verma'). None of the three registered authors (Rahul K, Manoranjan M, Karthikeyan J V) are in DataInitializer.java or LoginPage.jsx quick-fills. Must be updated before viva.

**[MISSING - HIGH] 2. Server-Tier Spring Security & Cryptographic JWT:** SecurityConfig.java permits all requests without token authentication (.permitAll()). Tokens are dummy strings ('jwt-token-campus-' + id) rather than signed RFC 7519 JWTs, and passwords are not checked with BCrypt hashes.

**[MISSING - MEDIUM] 3. Background Automated Deadline Alerts & Email Dispatch:** The report claims automated email alerts and review deadline schedulers. While in-app notification records exist, no SMTP JavaMailSender or Spring @Scheduled background tasks are present in the backend.

**[MISSING - LOW] 4. Physical Server Storage for Generated IDF Files:** The report claims generated IDFs are saved in a protected server directory with paths in DB. In reality, IDFs are generated on-the-fly inside the React PatentClaimsModal without creating physical disk files.

**[MISSING - LOW] 5. Retrospective Edit Lock on Weekly Logs:** The report claims restrictions prevent retrospective changes. In code, MiniProjectService allows students to re-submit and overwrite an existing log even after faculty approval.

## 3. WHAT IS EXTRA / BONUS (ADVANCED IMPLEMENTATIONS)

The codebase contains several high-value enterprise features that go well beyond the baseline descriptions in the project report. These provide substantial positive talking points during presentation:

[EXTRA / ADVANCED] **1. AI Faculty Mentor Recommendation Engine (Domain-Aware):** In ProjectCatalogueService.java, an intelligent staff matcher evaluates domain keywords against faculty departments and current workload (tracking active groups vs. max cap of 10) to recommend the optimal guide.

[EXTRA / ADVANCED] **2. Full-Fledged Academic Helpdesk & Ticketing Subsystem:** AcademicHelpdesk.jsx and AcademicQueryService.java provide a complete campus ticketing system where students submit formal queries on mentor allocations and domain requirements, complete with status tracking.

[EXTRA / ADVANCED] **3. Academic Catalogue Capacity Locking (Max 2 Students / Topic):** ProjectCatalogueService enforces strict student capacity quotas (max 2 students per topic), preventing duplicate bids and automatically updating remaining seat telemetry across departments.

[EXTRA / ADVANCED] **4. Multi-Deliverable Portfolio Tracking (GitHub, Figma, Demo, Video):** MiniProject.java & MiniProjectTracker.jsx allow students to attach live prototype URLs, Figma UI mockups, GitHub repository links, and video demo URLs, auto-advancing status to PROTOTYPE_SUBMITTED.

[EXTRA / ADVANCED] **5. Apache PDFBox Automated Proposal Text Stripper:** PdfTextExtractionService.java natively parses uploaded multi-page research PDFs on disk and feeds extracted text directly into the AI summarization and novelty analysis engine.

[EXTRA / ADVANCED] **6. Interactive Patent Studio: One-Click Print & Clipboard Dossier:** PatentClaimsModal.jsx features an integrated print engine (window.print()) formatted specifically for official IPO Form 2 layouts, complete with interactive copy buttons and toast notifications.

[EXTRA / ADVANCED] **7. Resilient Heuristic Fallback with 24 Indicator Keywords:** When Ollama is offline or times out (>4s), the backend never crashes; it seamlessly transitions to a 24-indicator lexical analysis engine that calculates novelty scores and IP classifications deterministically.

## 4. WHAT IS 100% VERIFIED & IMPLEMENTED (CORE BASELINE)

[DONE] React 19 & Vite 6 SPA running with 11 protected routes across 4 portals (Student, Faculty, Academic Office, IP Cell).

[DONE] Java 17 & Spring Boot 3.2.3 REST backend supporting both MySQL 8 and H2 in-memory databases with Dev profile.

[DONE] 12 JPA Relational Entities mapping users, proposals, projects, weekly logs, reviews, milestones, and IP filings.

[DONE] Weekly Progress Tracking with 16-week logging, technical hurdle documentation, and 5-star faculty rating modal.

[DONE] Automated IP Cell Handoff: Flagging a project as 'READY_FOR_IP' auto-initiates an IP filing with docket number.

[DONE] Local Ollama Llama 3.2 Integration with temperature=0.2 and 5-section patent evaluation structured prompt.

[DONE] Patent Claims Studio drafting Background, Independent Claim 1, Dependent Claims 2-5, and IPC Classification mapping.

## 5. EXAMINER STRATEGY & DEFENSE PLAYBOOK

**RECOMMENDED VIVA VOCE STRATEGY:**

1. Highlight the Extra Features: Present the Academic Helpdesk and AI Mentor Allocation as institutional enhancements designed to address real-world faculty workload constraints.
2. Demonstrate the Dual-Mode AI: Show that disconnecting Ollama seamlessly switches to the 24-indicator heuristic engine without server crashes—proving robust system fault tolerance.
3. Seed Actual Names: Update DataInitializer.java before 02.11.2026 so Rahul K, Manoranjan M, and Karthikeyan J V appear as the registered student inventors during the live demonstration.